Verifying Newton's Second Law with a Half Atwood Machine

SYNOPSIS

Newton's second law fits in five characters, but it makes a testable promise: double the net force on an object and its acceleration doubles; double the mass and the acceleration halves. This note puts that promise on trial with a cart on a track pulled by a hanging mass. The interesting part is that the obvious experiment *does not work* — and finding out why is most of the physics.

BEFORE YOU READ ON — PREDICT

The cart is pulled by a hanging mass and accelerates. You now **double the hanging mass**, which doubles the force pulling the system along.

Does the acceleration double?

Commit to yes or no, with one sentence of reasoning, before you turn the page.

PERFORMANCE EXPECTATIONS

1. **Draw** a free-body diagram for each body in a connected two-body system, and mark the positive direction of acceleration on each.
2. **Write** ΣF = ma for each body along its own direction of motion.
3. **Combine** the two equations to predict the system's acceleration, and explain why the total mass appears in the denominator.
4. **Explain** why adding mass to the hanger does not double the acceleration.
5. **Describe** a procedure that tests F = ma by changing the net force while holding the total mass fixed.

CARRY THIS INTO THE LAB

This note is Lab 4. The setup drawing, the two free-body diagrams and the derivation below are not background for that lab — they are its Introduction, its Experimental section and its analysis plan. You are expected to bring them into the report and build on them, section by section:

Introduction your own derivation of $a = F_{g,H}/M_T$, and the rearrangement $a = (1/M_T)F$ that turns the slope of a graph into a mass.

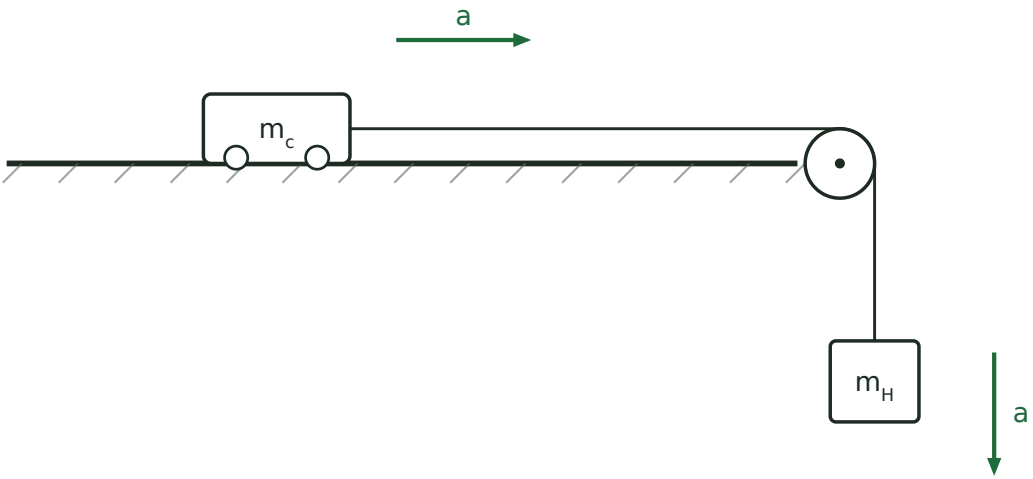
Experimental the setup drawing and both free-body diagrams, with the positive direction you chose for each body stated explicitly — and why total mass is held constant while the hanging mass increases.

Analysis the two things the lab has to confirm: that acceleration rises in proportion to force, and that the slope of that line is $1/M_T$.

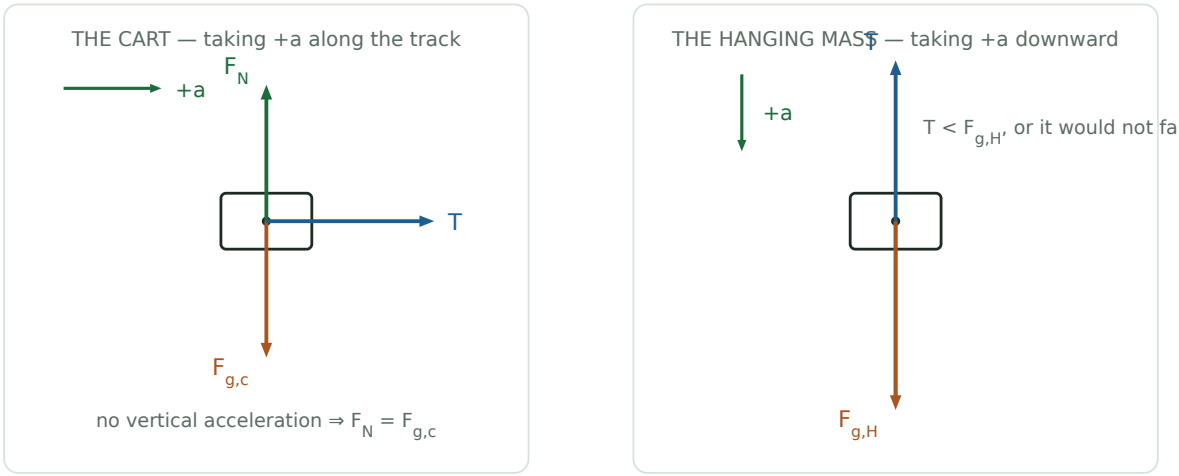
Discussion the three results below that your data must be checked against — the tension is less than the hanging weight, the acceleration is less than g, and one acceleration is shared by two bodies moving in different directions.

Lab 04 — Newton's Second Law →

The apparatus



Two bodies, two diagrams



1. Draw the setup in your guided notes — track, cart, pulley, string, hanging mass. Getting the geometry down is half of getting the physics right.

2. Draw both free-body diagrams, one per object, and mark +a on each.

3. Write Newton's second law for the **cart**, along the track.

4. Write Newton's second law for the **hanging mass**, taking down as positive.

Putting them together

Eq. 1 $T = m_c a$ — the cart — tension is its only horizontal force

Eq. 2 $F_{g,H} - T = m_H a$ — the hanging mass — gravity wins, but only by $m_H a$

Eq. 3 $a = \dfrac{F_{g,H}}{m_c + m_H}$

substitute the first into the second — the driving weight over the **total** mass

5. Substitute the cart equation into the hanging-mass equation and solve for a. Show every step.

Why your prediction was probably wrong

THE TRICK THAT MAKES THE EXPERIMENT WORK

Do not *add* mass to the hanger. **Move** it there from the cart.

Take a small mass Δ off the cart and hang it on the string instead. The system becomes (m_c − Δ) and (m_H + Δ). The **total** mass in the denominator has not changed by a gram — but the driving weight in the numerator has gone up. Now the only thing varying is the net force, which is precisely the experiment Newton's second law asks for.

6. Describe the verification strategy in your own words: what do you change, what do you hold fixed, what do you measure, and **what pattern must the data show** if F = ma is telling the truth?

7. Now design the **other** half of the test — the one that checks a ∝ 1/m. What would you vary this time, and what would you keep constant?

8. Run it. Transfer the mass in equal steps, record the acceleration each time, and plot a against the hanging weight. What shape should the graph be, and what does its slope represent?

BACK TO YOUR PREDICTION

You predicted whether doubling the hanging mass doubles the acceleration. Was it right? Point to the term in a = $F_{g,H}$ / (m_c + m_H) that your reasoning left out.

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